Unsupervised Classification

Unsupervised Classification

- **Commonly referred to as *clustering***
- **Unsupervised classification normally requires only a minimal amount of initial input from the analyst**
- **This is because clustering does not normally require training data**

Unsupervised Classification

- **Process that search for *natural groupings* of the pixels with similar spectral properties**
- **Clustering process results in a classification map consisting of *m* spectral classes (clusters)**
- **The analyst then uses prior *knowledge/* information to assign or transform the *spectral* classes into thematic *information* classes of interest (e.g., forest, agriculture)**

Unsupervised Classification

- **Some spectral clusters may be meaningless because they represent mixed classes of Earth surface materials (similar spectral response)**
- **The analyst must understand the spectral characteristics of the terrain (land covers and Uses) well enough to be able to label clusters as specific information classes**

Unsupervised Classification

- **Two step process**
  - **Iterative process of Clusters forming**
  - **Assignment of thematic classes to the clusters**

Unsupervised Classification -Methods

- **Two widely used unsupervised classification methods**
  - **K-Mean**
  - ***Iterative Self-Organizing Data Analysis Technique* (ISODATA)**

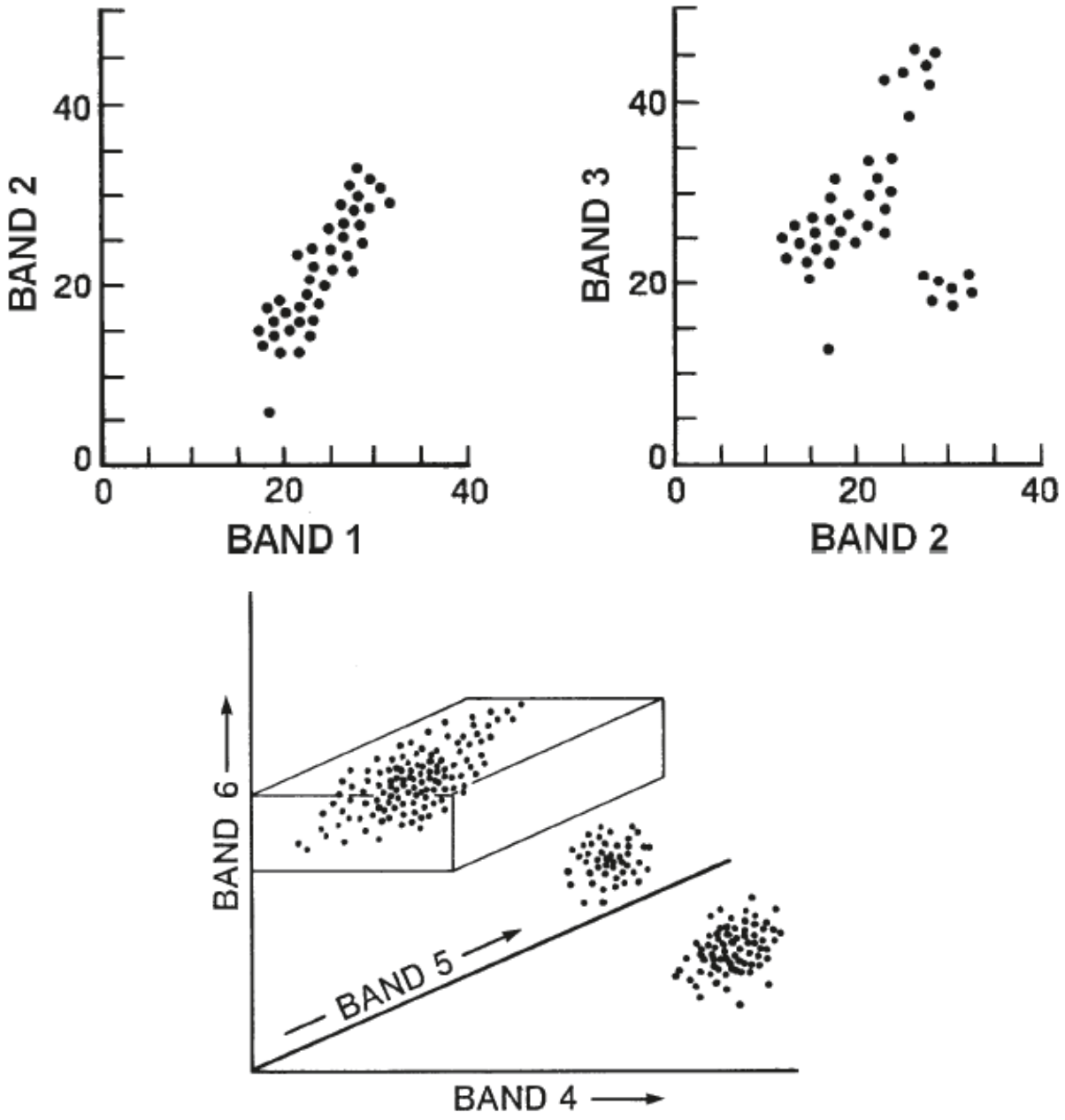
- **'K' in the name of the algorithm represents the number of groups/clusters we want to classify our data into**

K Means

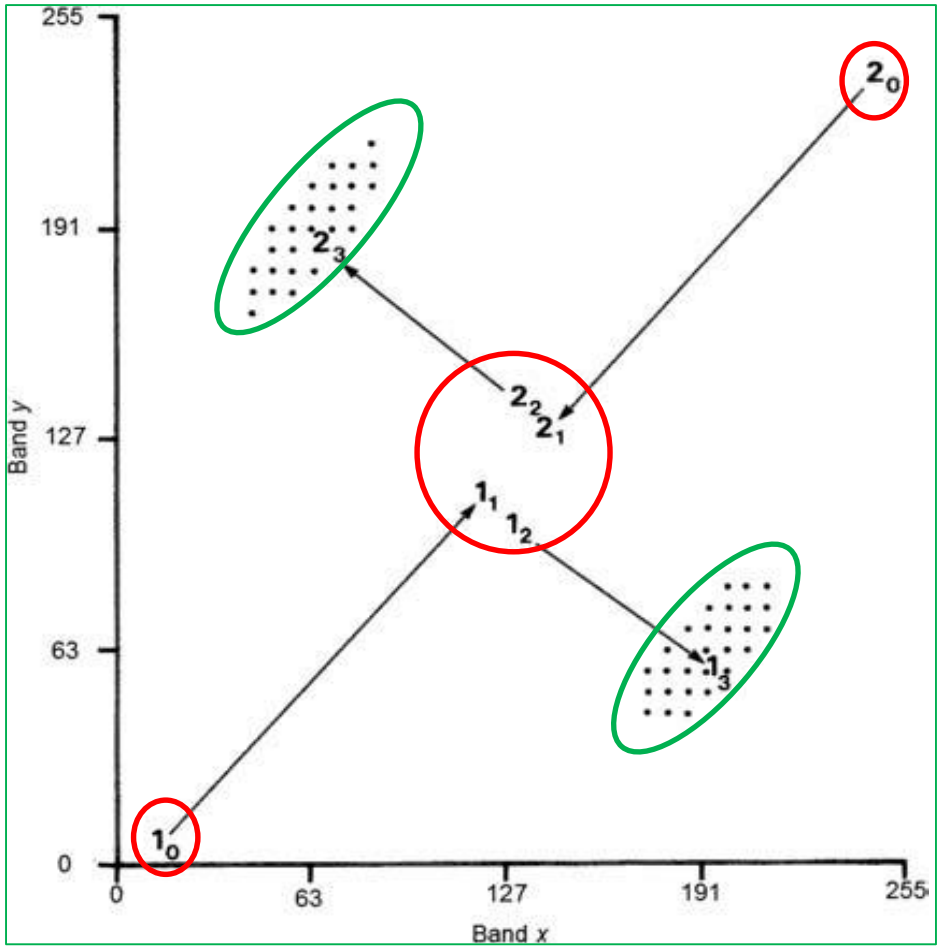
- K-Means Clustering is an Unsupervised Machine Learning algorithm, which groups the un-labeled dataset into different clusters
- The main idea of the clustering is to specify number of clusters (as a constraint) to be generated
- Clustering starts with the fixed a priori specified number of clusters to be created

K mean-Min Distance to Means

- **Uses the minimum spectral distance formula to form clusters**
- **Begins with either arbitrary cluster means or the means of an existing signature set**
- **Each time the clustering repeats, the means of these clusters are shifted**
- **The new cluster means are used for the next iteration**
- **It iteratively calculates the new cluster means and the pixels assignment is updated to the relevant clusters**
- **This iterative assignment process continues until there is no substantial change between the successive iterations**


BAND 2
40
20
0
0
20
40
BAND 1
BAND 3
40
20
0
0
20
40
BAND 2
BAND 6
BAND 5
BAND 4

- **Min distance to means**
  - **To classify unknown pixel, the distance between the unknown pixel and the mean values for each (cluster)class is determined**
  - **Assigns the unknown pixel to the closest class**

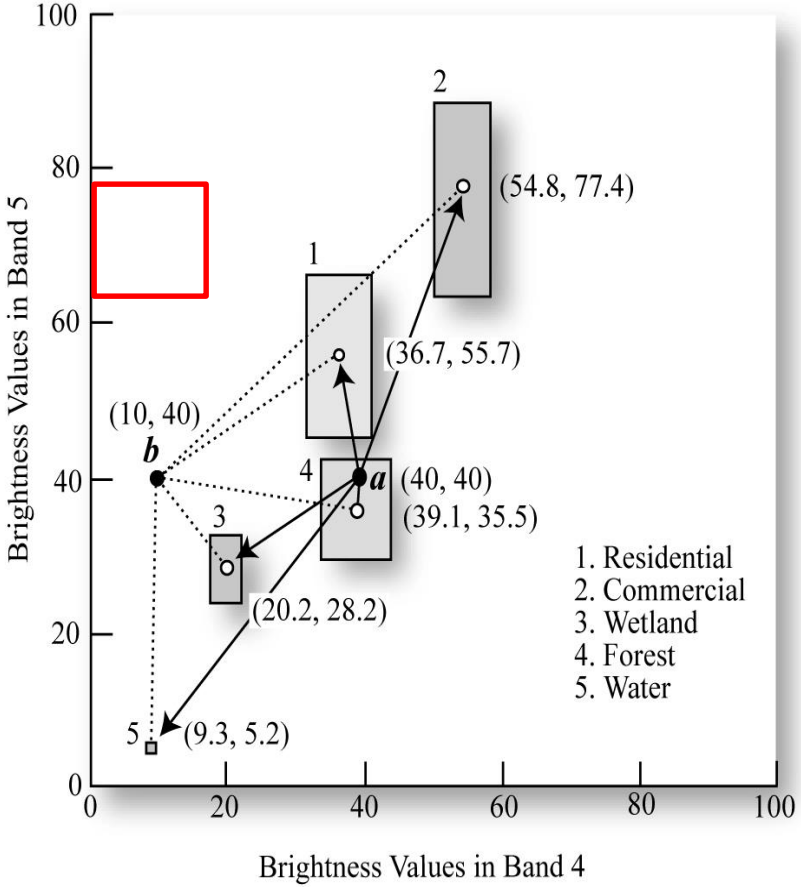


Minimum Distance to Means Classification Algorithm

- **The computation of the Euclidean distance from point to the mean of class 1 measured in bands k and L relies on the equation:**

$$Dist = \sqrt{\left(BV_{ijk} - \mu_{ck}\right)^2 + \left(BV_{ijl} - \mu_{cl}\right)^2}$$

- **where μ_{ck} and μ_{cl} represent the mean vectors for class *c* measured in bands *k* and *l***



Class	Distance from pixel *b* (10, 40) to the mean of each class
1. Residential	$\sqrt{(10-36.7)^2+(40-55.7)^2} = 30.97$
2. Commercial	$\sqrt{(10-54.8)^2+(40-77.4)^2} = 58.35$
3. Wetland	$\sqrt{(10-20.2)^2+(40-28.2)^2} = 15.75$ Assign pixel *b* to this class; it has the minimum distance
4. Forest	$\sqrt{(10-39.1)^2+(40-35.5)^2} = 29.45$
5. Water	$\sqrt{(10-9.3)^2+(40-5.2)^2} = 34.8$

ISODATA Clustering

- **The *ISODATA* algorithm is a modification of the *k*-means clustering algorithm, which includes**
  - **Merging of the clusters**
  - **Splitting of clusters**
- **ISODATA is iterative because it makes many passes through the remote sensing dataset until specified results are obtained**

ISODATA Clustering

- Uses the minimum spectral distance formula to form clusters
- The new cluster means are used for the next iteration
- The process stops when
  - a maximum number of iterations has been performed,
  - or a maximum percentage of unchanged pixels has been reached between two iterations

ISODATA Clustering

- **ISODATA algorithms normally require the analyst to specify the following criteria:**
  - **Maximum number of clusters (classes) to be identified by the algorithm**
  - **Minimum Distance: Cluster pairs that have a Euclidean distance less than this value are merged into one cluster**
  - **No of iterations-maximum number of times ISODATA is to classify pixels and recalculate cluster mean vectors. The ISODATA algorithm terminates when this number is reached**

ISODATA Clustering

- **Minimum members in a cluster (%): If a cluster contains less than the minimum percentage of pixels, it is deleted, and the members are assigned to an alternative cluster (percent of pixels in a cluster relative to the total number of pixels in the image)**

- ***Maximum standard deviation*** **(σ_{max}): When the standard deviation for a cluster exceeds the specified maximum standard deviation in any band, the cluster is split into two clusters**

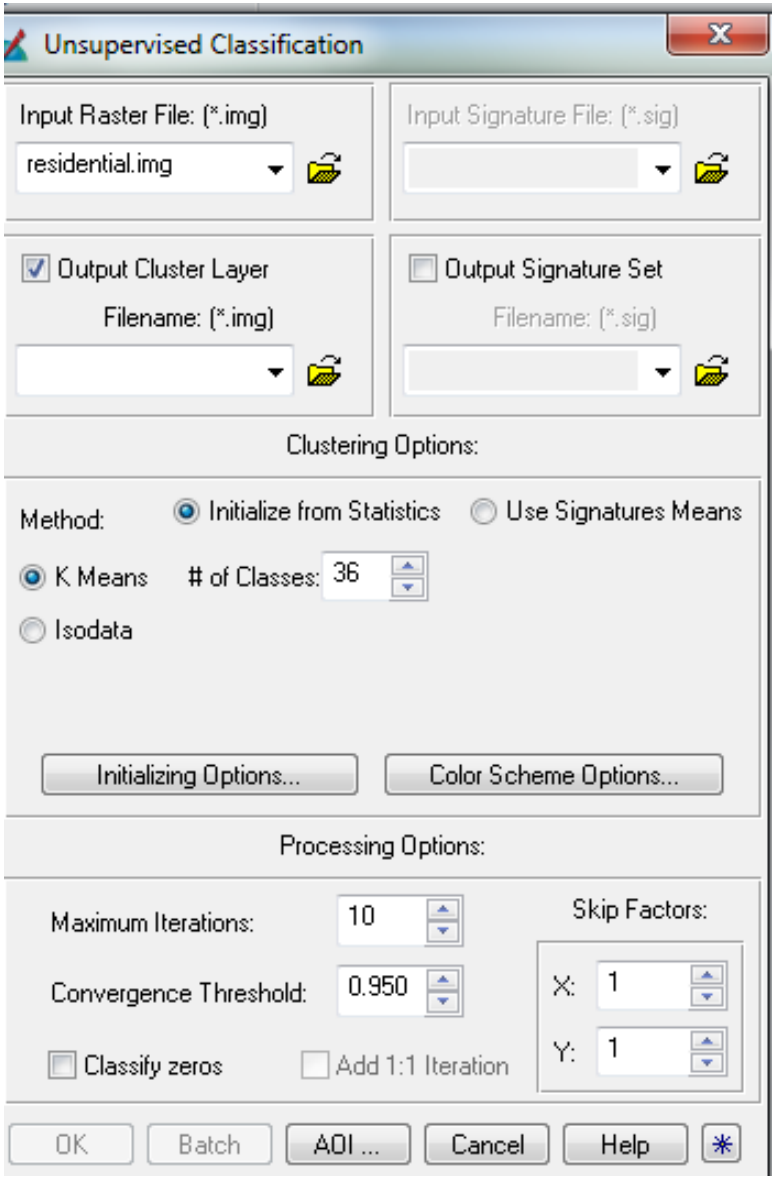
ISODATA
(*Iterative Self-Organizing Data Analysis Techniques)*

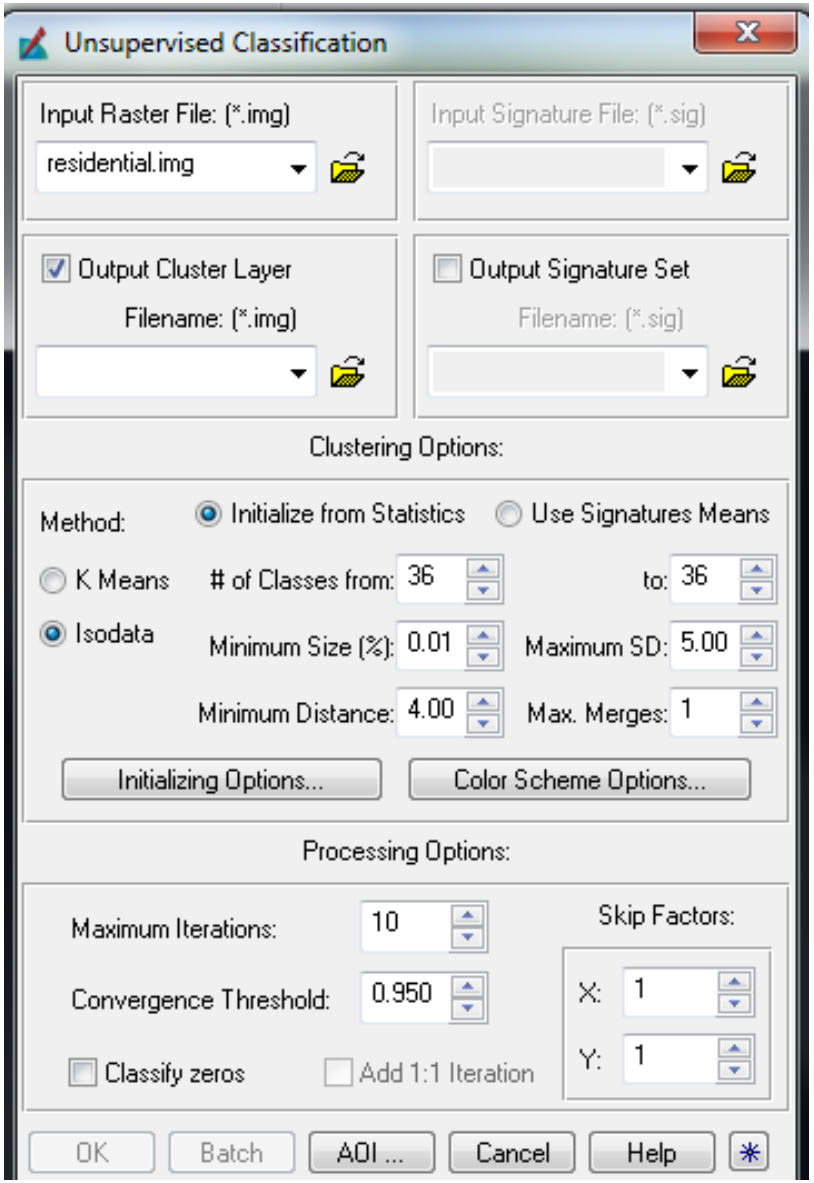
- **It additionally merges the clusters that have a smaller number of pixels than the threshold or if the center of two clusters are too close**
- **It also split a big cluster if the cluster standard deviation exceeds a predefined value**
- **After the required number of clusters are formed, the analyst using his prior knowledge about the area, or some reference data to assigns the information classes to each cluster to prepare the final thematic map**

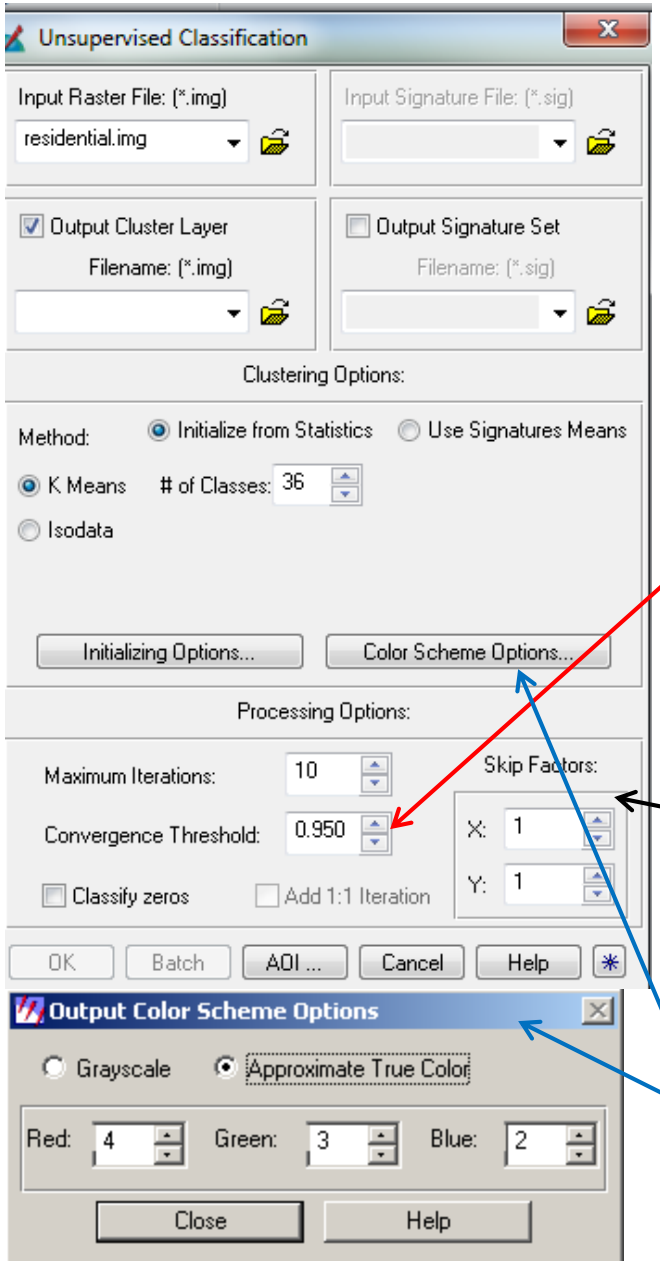
Post Classification

- In post classification phase, analyst compare spectral classes with some reference data to identify the spectral classes
- Spectral reflectance curves can be used to identify the spectral classes
- Defining the level of classification
- Merging different classes to reach outcome
- Accuracy assessment through field truthing or reference data

Unsupervised Classification







Initialize from Statistics generate arbitrary clusters from the file statistics for the entire .img file

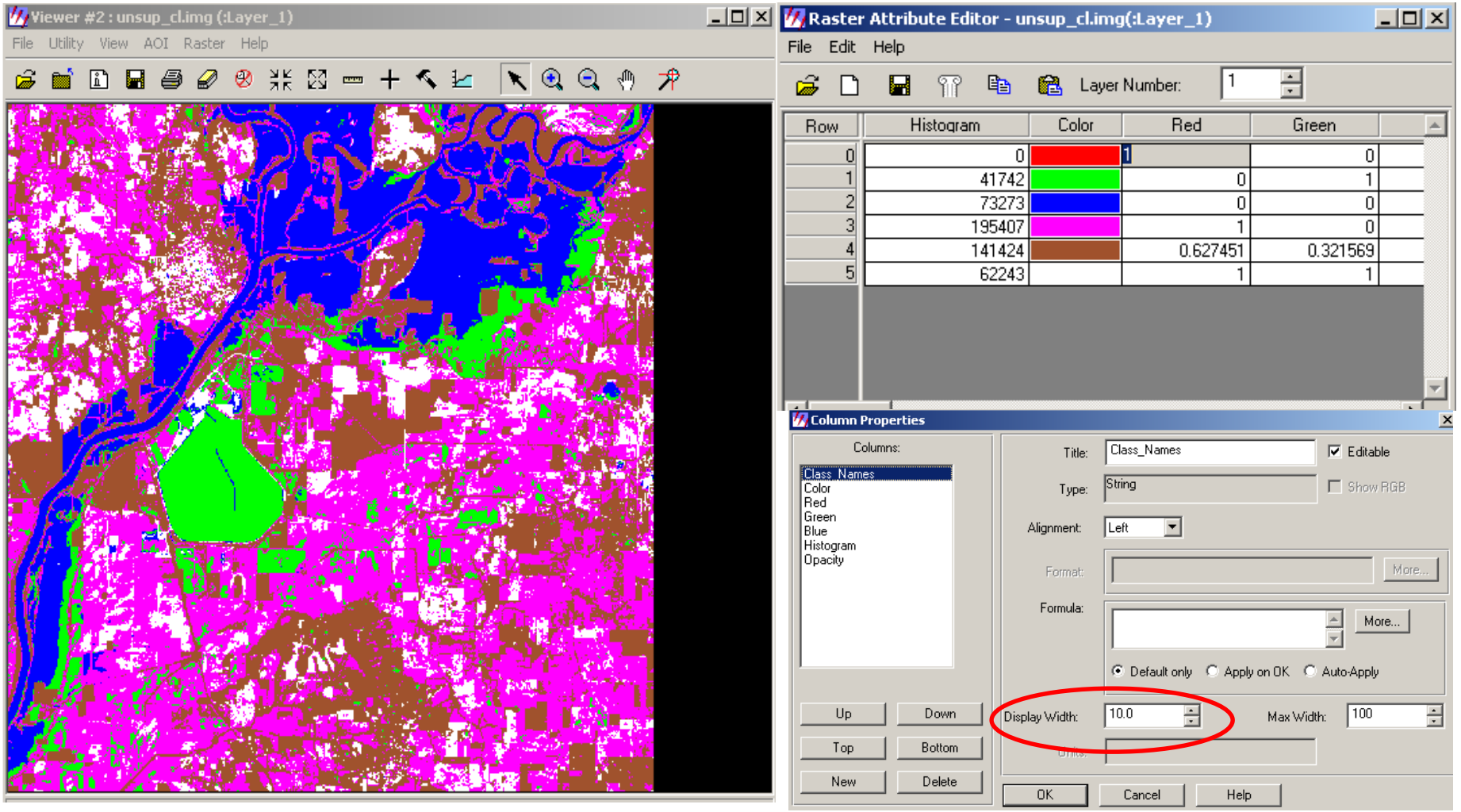
Use Signature Means use only the selected signatures in the Signature Editor to generate the clusters

Convergence Threshold: is the maximum percentage of pixels whose cluster assignments can go unchanged between iterations

Skip Factors: X,Y:
1 processes all pixels,
2 processes every other pixel,
3 every third pixel

Assignment of color to the clusters

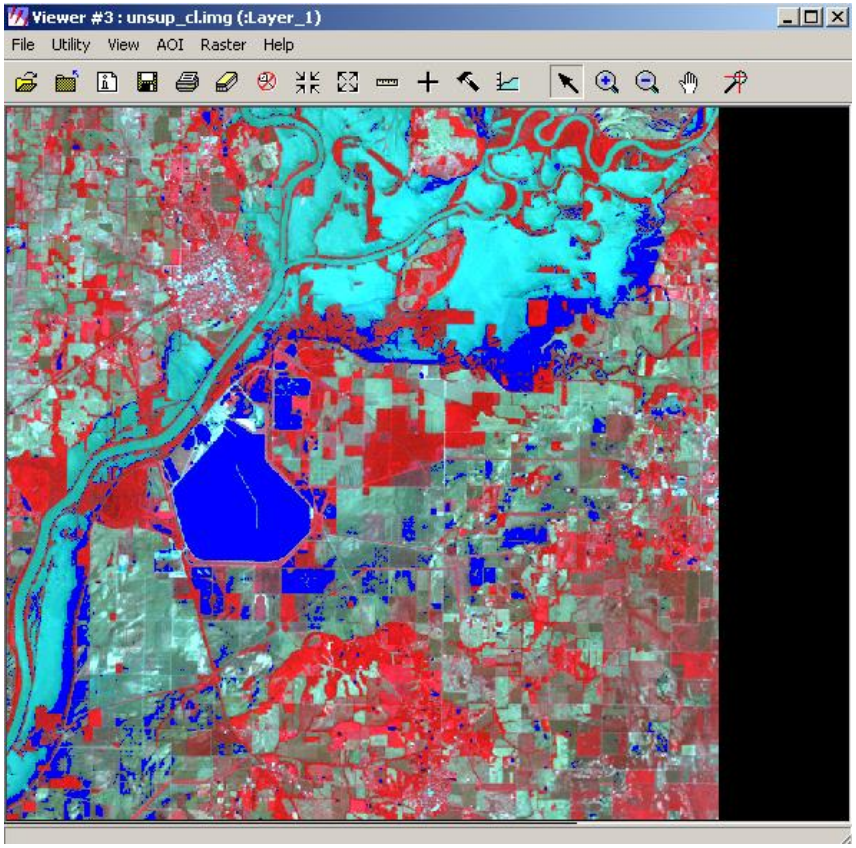
Classified Image

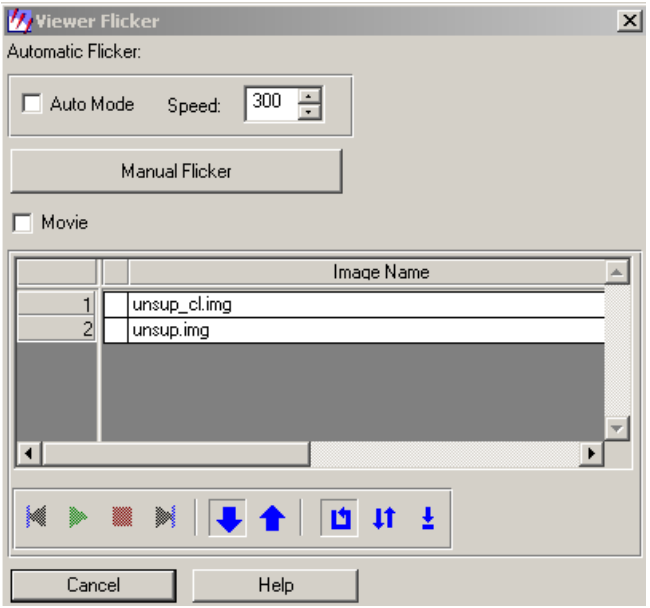


Raster Attribute editor > edits> column properties > up /down > Display width

Cluster Assignment to classes

Display both original and classified image in viewer





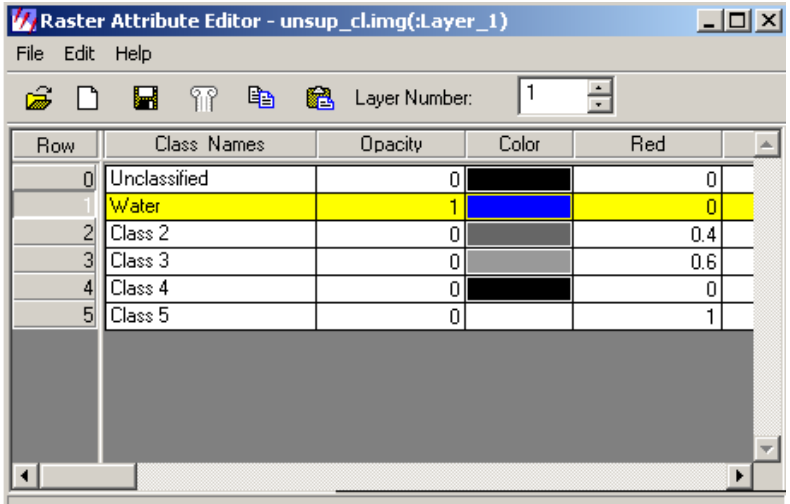


Figure 1. Washington DC Mall data set.

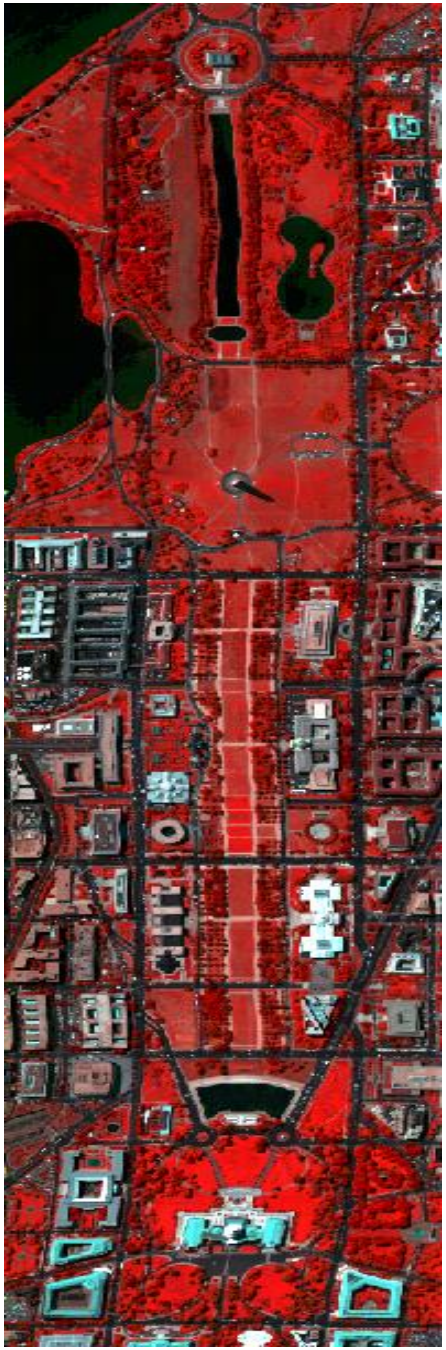
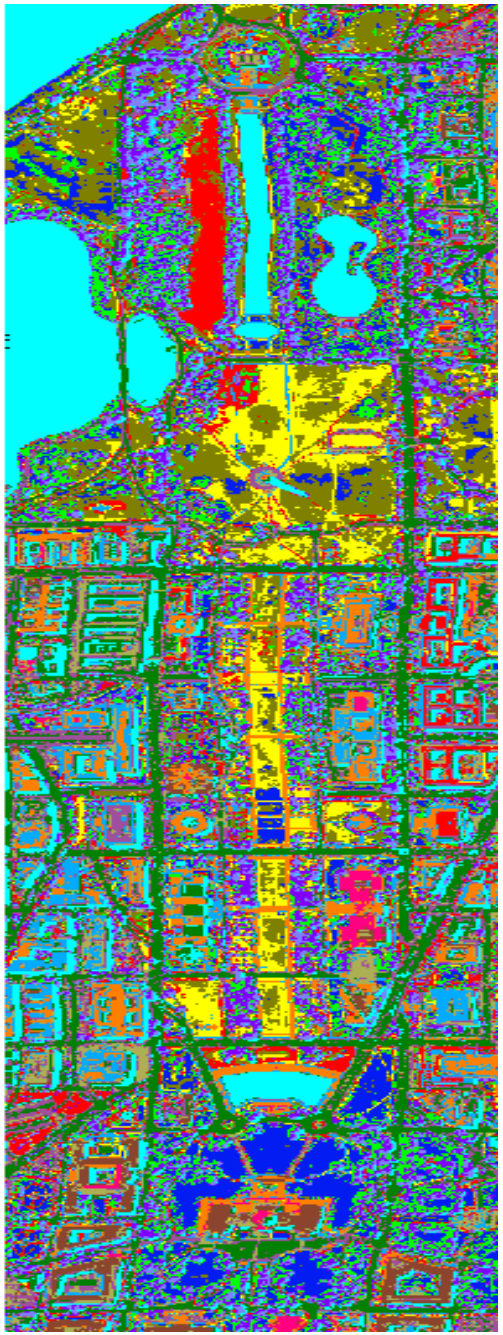
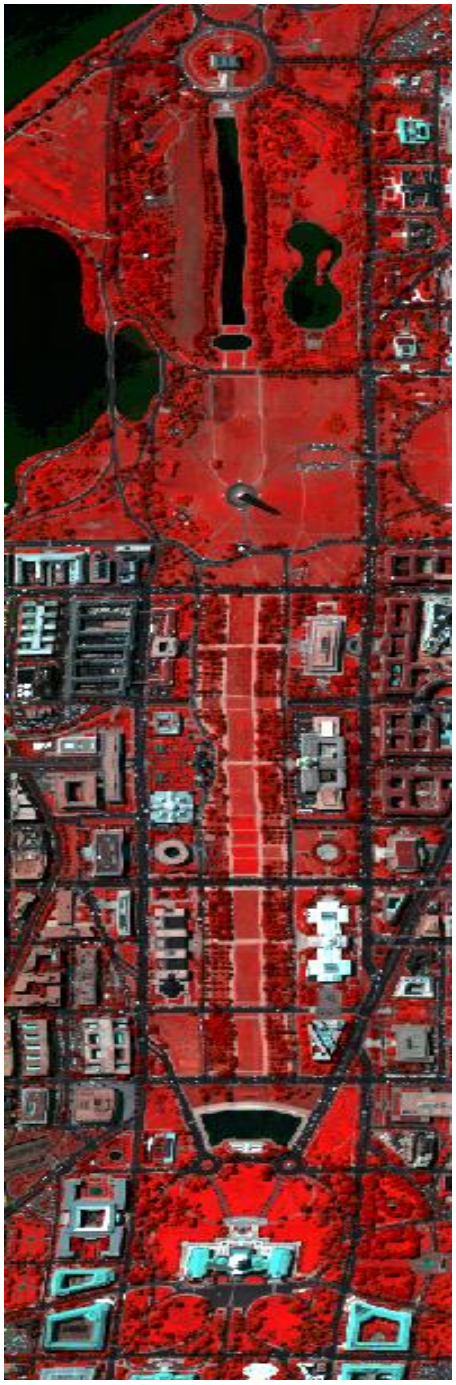
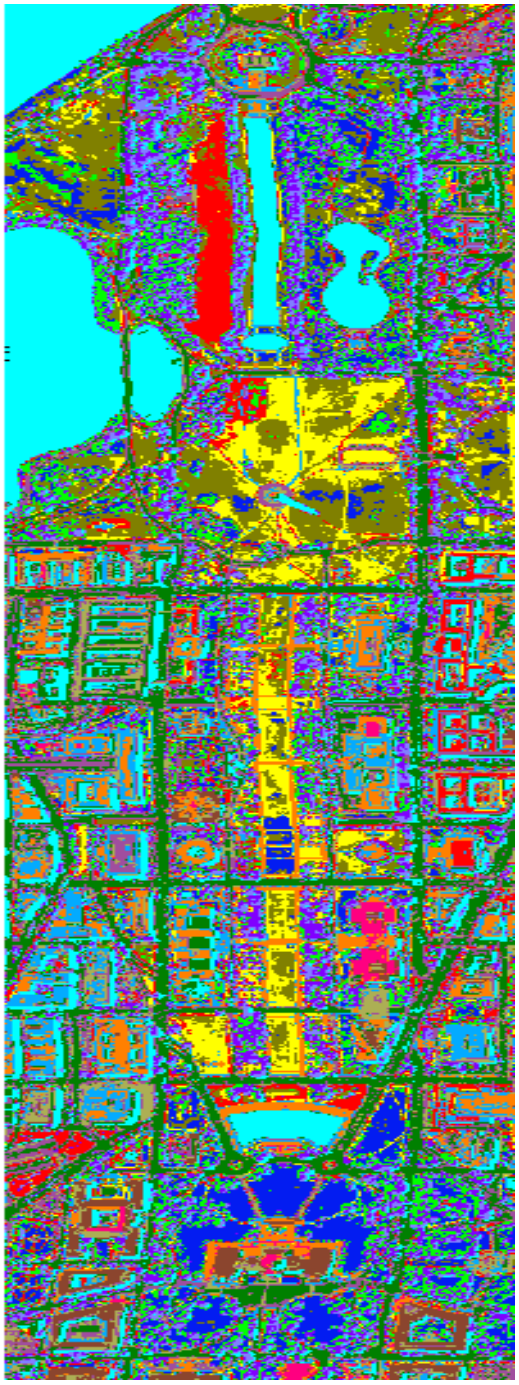
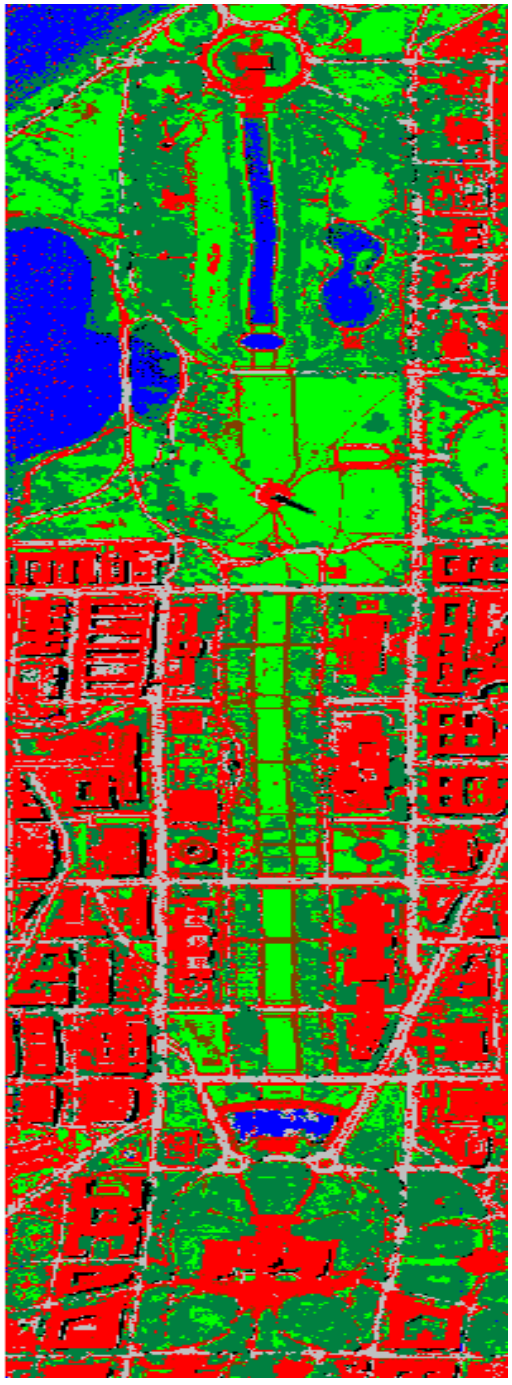


Figure 2. ISODATA cluster map of the Washington DC Mall. A total of 15 clusters were identified in order to discern the spectral heterogeneity of information classes.









Classes

- Roof
- Street
- Path
- Grass
- Trees
- Water
- Shadow

Reference/Reading

- Chapter 12,Section 12.3, Campbell
- Chapter 8, Section 8.3.2, Mather